Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
B		2527943

Tourists or Icebergs？Dynamic Optimization of Juneau's Tourism System under Feedback Constraints

Summary

In this study, a **sustainable tourism model** system based on an **optimization problem** is developed to maximize comprehensive revenue from tourism while considering the environment,the carrying capacity of infrastructure and residents' satisfaction, promoting long-term sustainable development.

A **multi-objective optimization model** is created using **system dynamics** for key factors affecting tourism revenue in Juneau City. **Differential equations** are used to quantify dynamic relationships among tourist growth, glacier melt rate, environmental carrying capacity, and resident satisfaction. The optimization problem aims to **maximize comprehensive revenue** with the number of tourists and tax rate as decision variables, subject to constraints such as environmental pressure, infrastructure stress, and satisfaction thresholds. A **genetic algorithm** is applied to solve the model, determining the optimal equilibrium point and tax allocation for maximizing tourism benefits within environmental and satisfaction limits. Results show that the optimal number of tourists for sustainable tourism in Juneau is **3,762 per day**, with an optimal tax increase of **1.6%**. Additionally, an **economic-ecological feedback mechanism** is integrated, and **Sobol sensitivity analysis** identifies key factors for model optimization.

To assess system compatibility, **cluster analysis** is used to categorize cities based on **ecological sensitivity (E), cultural preservation (C), and infrastructure capacity (I),** identifying ecologically fragile, infrastructure-deficient, and culturally conflicting cities. The model can be **adjusted adaptively** for these categories, with Mumbai used as an example of a city with insufficient infrastructure. By combining attractiveness equations, flow control algorithms, and investment feedback mechanisms, an optimized strategy was developed to balance tourist numbers and distribute resources fairly, enhancing **model practicality**.

Finally, a **memo** summarizes the core predictions and the impact of each measure, proposing an optimization scheme. This **multi-objective optimization sustainable tourism model** offers a tool for quantitatively assessing and optimizing tourism development. It effectively balances multiple objectives, maximizes benefits, and is adaptable to different cities, providing a reference for sustainable development.

Key words: Sustainable tourism models; multi-objective optimization; feedback mechanisms;

system dynamics

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I. Introduction

1.1 Background

With the rapid development of global tourism, the impact of tourism on the environment, economy and society is becoming increasingly apparent. Juneau, the capital of Alaska, USA, is a popular destination for tourists because of its unique and magnificent glacier mountains and rich cultural history. However, with the surge in the number of tourists, especially cruise ship tourists, Juneau's natural resources and local society are facing corresponding challenges.

Fig1.Juneau, Alaska

Sustainable tourism, as an emerging tourism development model, emphasizes meeting the current needs of tourists while minimizing the negative impact on the natural environment, cultural heritage and local communities, so as to ensure the long-term healthy development of the tourism industry. From the perspective of sustainable tourism, the development of tourism in Juneau needs to focus on ecological protection, environmental management and balancing the interests of the community in order to ensure the long-term healthy and balanced development of the tourism industry.

1.2 Restatement of the Problem

• Establish a sustainable tourism model that balances the number of tourists and environmental pressures while ensuring resident satisfaction, and maximizes tourism revenue.

•Identify optimization factors and constraints, and establish a feedback mechanism between different factors.

• Conduct a sensitivity analysis of the above factors to reflect the degree of influence of factors on the sustainable tourism model.

• Apply the established sustainable tourism model to other tourist destinations affected by overtourism, and analyze using the model as an example to balance the local tourism industry with constraints.

1.3 Our works

To sum up the full article, we

• develop a sustainable tourism model based on multi-objective optimization to balance economic benefits, environmental pressure, infrastructure pressure and resident satisfaction. Starting from small models of each core factor, a target function of comprehensive benefits is constructed, which

means that under the premise of ensuring environmental sustainability, not overloading urban infrastructure and good resident satisfaction, the long-term comprehensive tourism benefits of Juneau are maximized by dynamically adjusting taxes and fees, setting limits on the number of tourists and optimizing capital allocation, and the constrained conditions are solved.

• analyze many influencing factors and construct“economic-ecological-infrastructure-social satisfaction feedback mechanism”to reflect the interaction between many influencing factors.

• conduct sensitivity analysis on sustainable tourism to reflect the degree of influence of each core factor on the sustainable tourism model.

• establish a classification model for tourist destinations, and then optimize the established sustainable tourism model to make it have good adaptability, be able to reasonably analyze other classified tourist destinations, and give reasonable measures, so as to achieve the sustainable development of urban tourism and the dynamic balance between tourist attractions in tourist destinations.

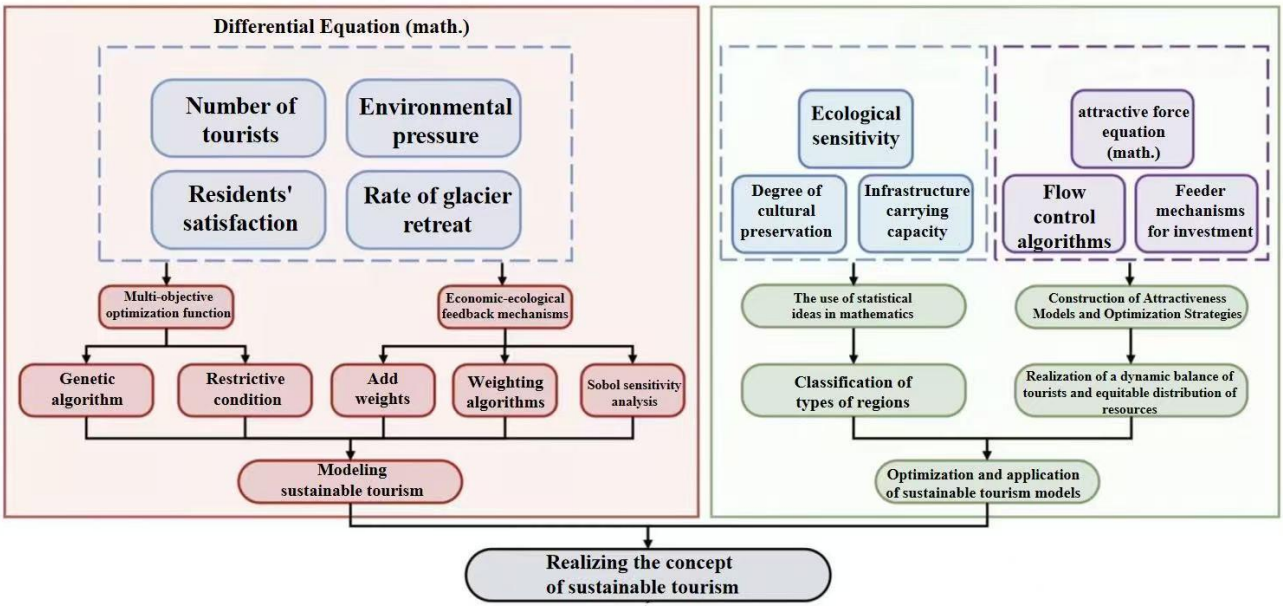


Fig2. Flowchart of the whole scheme

II. Models Preparations

2.1 Assumptions

• **Economic and social impact assumptions:** Tourist spending and tax revenue are important sources of income for cities; social satisfaction is affected both positively and negatively by employment opportunities and local congestion brought about by tourism.

• **Policy regulation assumption:** The government can regulate the number of tourists and environmental pressures by adjusting the tax and fee and investment ratios; tax revenue is only used for environmental construction, infrastructure construction and social and resident satisfaction construction projects.

• **Data integrity assumption:** Each city's indicators (such as glacier coverage, endangered species density, carbon emission intensity, etc.) can be obtained.

• **Indicator independence assumption:** The indicators are independent of each other and there is no

significant cross-impact.

• **Standardization processing assumption:** Indicators of different dimensions can be compared by standardization processing.

2.2 Data collection

Search a large number of authoritative official US databases, such as the US Open Database, to derive a small database for this question, and then perform subsequent data analysis for modeling.

III. Notations

Symbol	Definition
t	Time (unit: year)
$N(t)$	Number of tourists at time *t*
N_0	Without reform Initial number of tourists
μ	Tourist growth rate
$T(t)$	Tax revenue
$R(t)$	Environmental protection investment
γ	Tax impact on tourist number
$E(t)$	Additional environmental pressure at time *t*
$I(t)$	Infrastructure pressure at time *t*
$S(t)$	Social and residential dissatisfaction at time *t*
κ	Constant factor for glacier melting
I_{param}	Disturbance composite index

IV. Models

4.1 Establishment of a sustainable tourism model

In order to establish a sustainable tourism model for the city of Juneau, we need to identify the factors that affect the development of sustainable tourism, which can be divided into four parts :

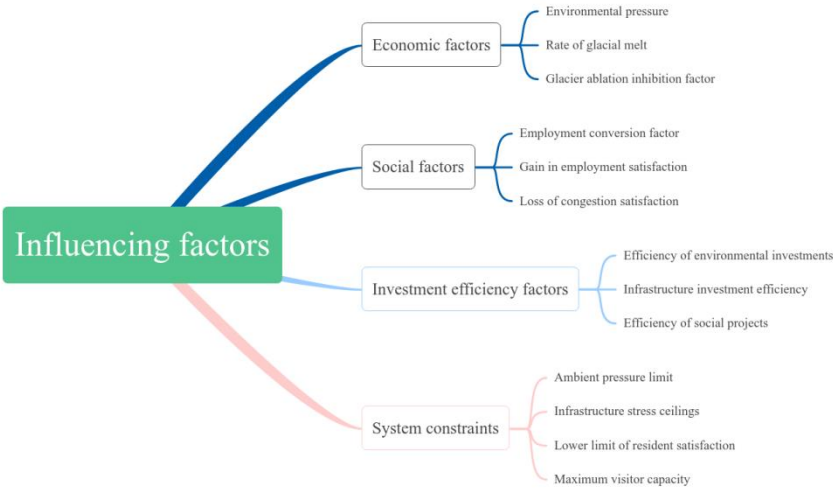


Fig3. Influencing factors

Based on the above factors and the preparation of preliminary data, we have extracted the most influential factors—tourist numbers, the rate of glacier melt, environmental pressure and resident satisfaction — and established corresponding models for each. Combining this with the core development concept of sustainable tourism and the corresponding constraints, the question of "maximizing tourism revenue while balancing tourist numbers and environmental pressure while ensuring resident satisfaction" is finally resolved. Then, by combining the analysis of the existing data, it can be concluded that all four leading factors are strongly correlated with time, so t is used as a variable. At the same time, the many influencing factors above interact with the main influencing factors to form a feedback mechanism. Finally, weights are set to rank a series of key parameters and sensitivity analysis is performed to reveal which factors are indispensable for sustainable tourism.

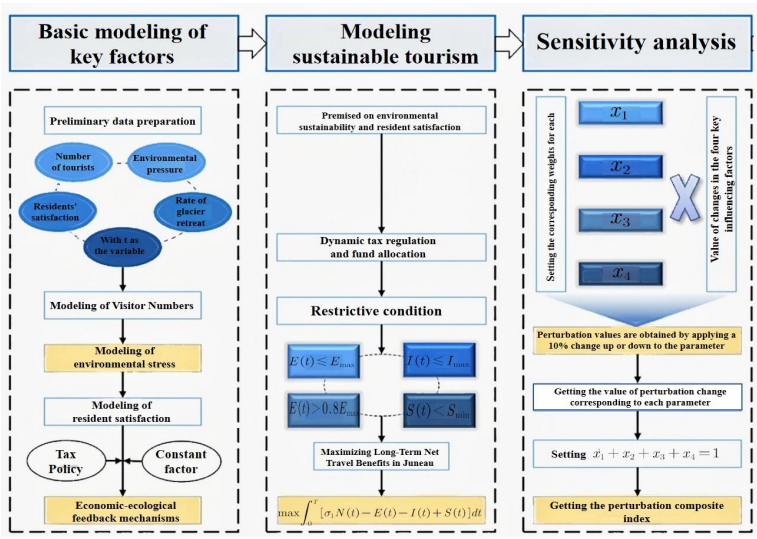


Fig4.Flowchart for question one

4.1.1 Modeling of the number of tourists

The number of tourists is one of the core variables in the model. We set $N(t)$ as the number of tourists visiting Juneau each year. On the one hand, an increase in the number of tourists directly increases their tourism revenue, but it may also increase the pressure on the operation of local infrastructure and impose a heavy environmental burden. At the same time, the carbon emissions

caused by tourists will accelerate the melting of glaciers, reduce the attractiveness of the local area to tourists, and reduce economic benefits as the number of tourists decreases. On the other hand, in order to address environmental degradation and infrastructure overload, we impose a tourism tax on each tourist. The revenue from the tax is invested in environmental protection, infrastructure construction and local social development. An increase in the tax will have a positive effect on local environmental governance, infrastructure improvement and resident satisfaction, but it will also directly reduce the number of tourists. In short, the number of tourists is usually affected by a combination of factors, including the environment, infrastructure and tourist taxes of the tourist destination. They are interrelated, and the specific equation (4-1) is as follows:

$$\frac{dN(t)}{dt} = N(t)u_0 - \gamma T(t)N(t) - \mu G(t)N(t) \tag{4-1}$$

Where，$N(t)$ is the number of tourists visiting the city of Juneau each year and u_0 is the coefficient of proportionality of the number of tourists growing naturally, so $N(t)u_0$denotes natural growth, and similarly, $\gamma T(t)N(t)$ denotes tax and fee suppression, and $\mu G(t)N(t)$ denotes glacial retreat suppression. Reflects that natural growth is inhibited by a combination of tax and fee policies and glacier retreat.

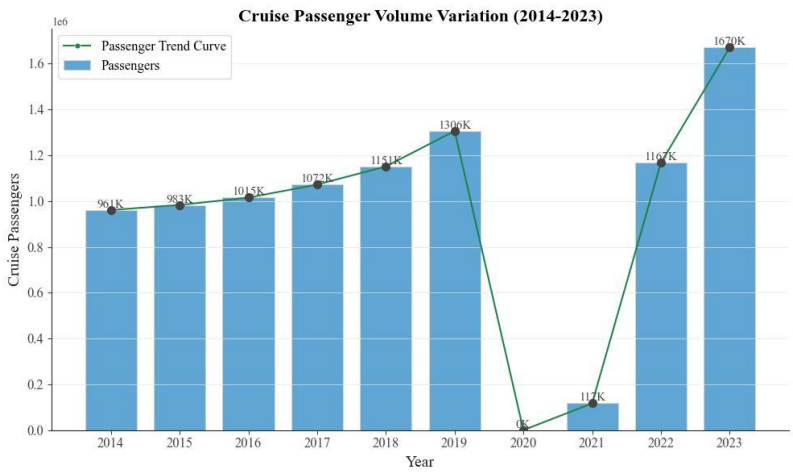


Fig5.Cruise passenger volume variation(2014-2023)

4.1.2 Modeling of environmental stress

The environmental impact of over-tourism is a critical issue for the City of Juneau. Tourist activities exacerbate the destruction of natural attractions, ecological degradation, and consumption of energy and water resources; glacier melting is exacerbated by the pressure of carbon emissions from tourists, which threatens the ecological balance; and increased investment in environmental protection and reduction of carbon emissions is the key to mitigating these problems, as represented by Equation (4-2) below:

$$E(t) = \sigma_2 N(t) + \kappa N(t) - \beta_1 \alpha_1 R(t) \tag{4-2}$$

Where, $E(t)$is the additional environmental pressure at the$t$ moment, $\sigma_2$is the average environmental burden coefficient per tourist, so $\sigma_2 N(t)$ is expressed as the direct pressure of tourists, similarly, $\kappa N(t)$is expressed as the glacier ablation contribution, $\alpha_1$is expressed as the proportion of taxes and fees spent on the environment, $\beta_1$is expressed as an efficiency coefficient of the use of taxes and fees for the

environment and $\beta_1\alpha_1 R(t)$is expressed as the environmental protection investment mitigation. A triple source of environmental pressure is revealed, i.e., tourist activity, glacier ablation, and environmental investment.

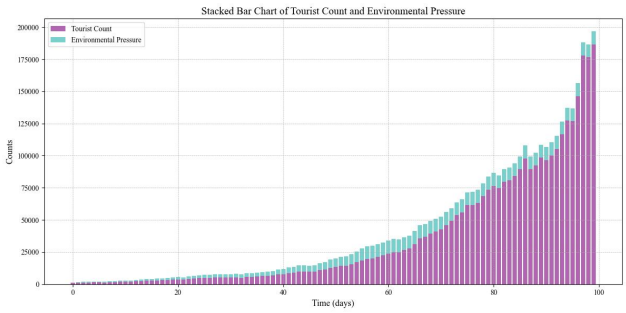


Fig6.Stacked bar chart of tourist count and environmental pressure

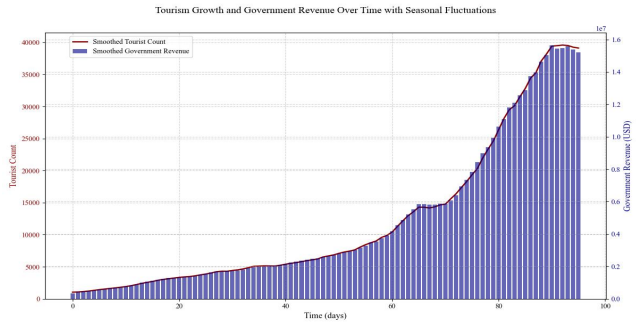


Fig7.Tourism growth and government revenue over time with seasonal fluctuations

4.1.3 Modeling of resident satisfaction

The development of sustainable tourism cannot be achieved without taking into account the satisfaction of the population. While the increased employment opportunities generated by tourism contribute to economic development, the accompanying traffic congestion and infrastructure pressures can reduce the quality of life of residents. In addition, the gain of social programs can enhance residents' happiness and satisfaction, especially with the financial support from tourism revenue. The specific equation (4-3) is expressed as follows:

$$\frac{dS(t)}{dt} = (\epsilon\lambda - \eta)N(t) + \beta_3\alpha_3 R(t) \tag{4-3}$$

Where $S(t)$is the moment's resident satisfaction, $\epsilon\lambda$ is the coefficient of employment increase, η is the coefficient of congestion brought by tourism, so $(\epsilon\lambda - \eta)N(t)$ expressed as the net effect of employment-congestion, the same way, $\beta_3$expressed as the tax and fee for the construction of social residents' satisfaction, α_3 expressed as the tax and fee for the construction of social residents' satisfaction, $\beta_3\alpha_3 R(t)$ expressed as the social program gain. The overall balance is between employment gains and congestion losses from tourism.

4.1.4 Modeling of infrastructure stress

The growth of the number of tourists is simulated based on an exponential growth model. The number of tourists at each time step is influenced by the number of tourists at the previous moment and a certain amount of random noise is taken into account. At the same time, the infrastructure pressure is proportional to the number of tourists, which is expressed through the parameter β (constant of proportionality). In addition, the infrastructure stress model takes into account the effects of noise over time.

$$T(t) = T(t-1) \times (1 + \text{growth rate} + \epsilon) \tag{4-4}$$

$$P(t) = \beta \times T(t) + \epsilon \tag{4-5}$$

In Equation(4-4), ϵ is the standard normally distributed noise, which represents the random fluctuations in the actual situation.In Equation (4-5),ϵ is again time-dependent noise, used to model

randomness in facility use.

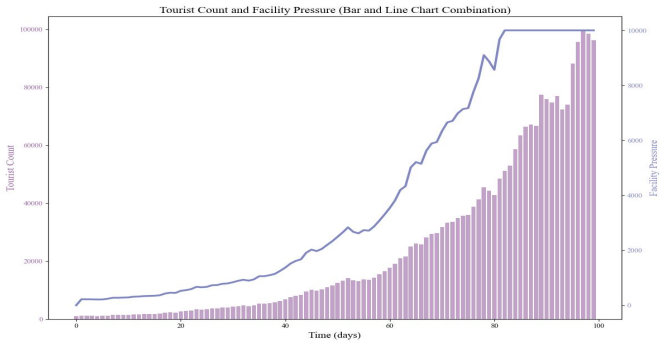


Fig8.Tourist count and facility pressure (bar and line chart combination)

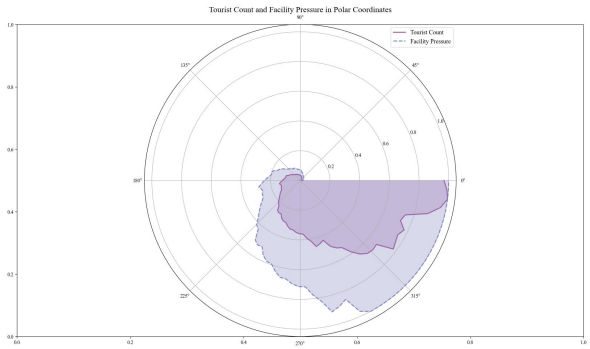


Fig9.Tourist count and facility pressure in polar coordinates

4.1.5 Feedback mechanisms

As the number of tourists grows, carbon emissions from transportation, lodging, and other tourism activities increase significantly, further exacerbating greenhouse gas emissions. These emissions accelerate climate change, leading to the accelerated melting of glaciers, which in turn threatens the natural landscape and ecosystems of the City of Juneau, as expressed in equation (4-6)as follows.

$$G(t) = \int_0^t \kappa N(\tau) d\tau \qquad (4\text{-}6)$$

Where $G(t)$is the amount of melting or retreat of the glacier over t. $\kappa$is a constant factor that may be related to the sensitivity of the glacier to external factors (such as carbon emissions), and $N(\tau)$is the number of tourists at the momentτ. It can be seen that an increase in tourists leads to an increase in carbon emissions, which directly leads to the melting of glaciers.

At the same time, the growth of tourists $T(t)$ is directly regulated by tax policies ($\gamma T(t)$) and ecologically inhibited by glacial melting $G(t)$ ($\mu G(t)$). The amount of glacial melting $G(t) = \int_0^t \kappa N(\tau) d\tau$ reflects the cumulative environmental impact of tourists' carbon emissions, forming a closed-loop negative feedback, as shown in the figure:

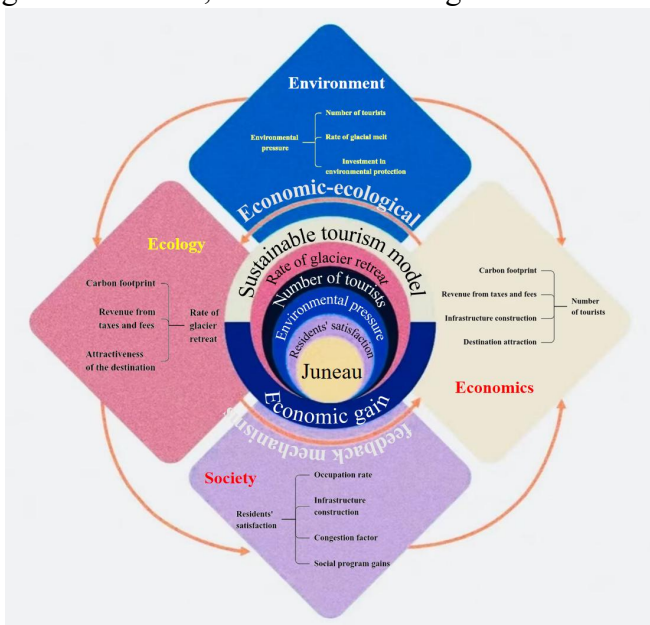


Fig10.Feedback mechanisms

The feedback mechanism in the figure above shows that an increase in tourists will lead to an

increase in carbon emissions, which will accelerate the melting of glaciers. The melting of glaciers will in turn have a negative impact on tourists, ultimately leading to a decrease in the number of tourists. At the same time, an increase in the number of tourists will drive the development of the tourism industry, leading to an increase in employment opportunities and higher levels of satisfaction among residents. However, it will also cause a series of problems such as traffic congestion, which will lead to lower levels of satisfaction among residents.

4.1.6 Constraint

In order to ensure the feasibility of a sustainable tourism model, we have set the following constraints based on practical considerations and data analysis:

- **Environmental protection constraints:** environmental pressures $E(t)$ must not exceed the set maximumE_{max}, i.e.:

$$E(t) \leqslant E_{max} \tag{4-7}$$

This constraint ensures that the environment will not become overburdened by too many tourists, avoiding ecological damage and protecting the natural landscape.

- **Infrastructure carrying capacity constraint:** The pressure on infrastructure $I(t)$ cannot exceed the maximum carrying capacity $Imax$, i.e.:

$$I(t) \leq I_{max} \tag{4-8}$$

This constraint ensures that the infrastructure can effectively support the needs of tourists while avoiding problems such as traffic jams, power shortages, and insufficient water resources.

- **Dynamic regulation:** Automatically increase taxes and fees at the time $E(t) > 0.8E_{\max}$;
  Reduce taxes and fees at the time $S(t) < S_{\min}$ to stimulate employment.

4.1.7 Establishing the goal equation for sustainable tourism

Based on the above model, a comprehensive objective equation (4-9) is established, that is, to ensure environmental sustainability and resident satisfaction, and to maximize Juno's long-term comprehensive tourism revenue through dynamic tax regulation and fund allocation.

$$\max \int_0^T [\sigma_1 N(t) - E(t) - I(t) + S(t)]\, dt \tag{4-9}$$

Among them, there are economic benefits $\sigma_1 N(t)$, environmental costs $E(t)$, infrastructure construction costs $I(t)$, and social benefits $S(t)$.

At the same time, the constraints are satisfied:a.$E(t) \leqslant E_{max}$

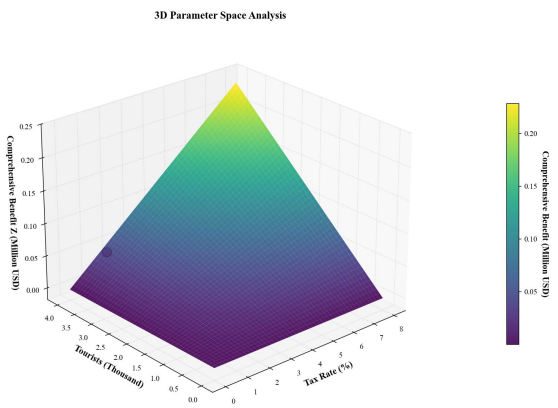
b.$I(t) \leq I_{max}$

c.$E(t) > 0.8E_{\max}$

d.$S(t) < S_{\min}$

4.1.8 Final optimization result

Based on the establishment of the final objective function, the following conclusions are drawn: the optimal number of tourists is controlled at 3,762 per day, and tax revenue is controlled at around 1.6%, as shown in the figure below :



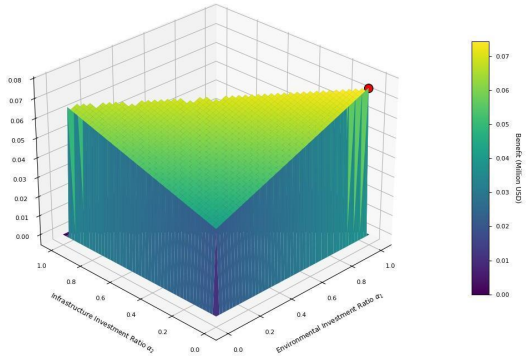


Fig11. 3D parameter space analysis-1 **Fig12. 3D parameter space analysis-2**

Fig11 is a three-dimensional plot of the relationship between the combined revenue and the number of visitors - tax rate, which is a good visualization of the different combined revenues for different numbers of visitors and tax rates. The optimal tax rate is 1.6% and the optimal number of visitors is 3762.Fig12 reflects the different combined returns for different investment ratios (environmental investment - infrastructure investment - investment in the satisfaction of the social population) after both the fixed tax rate and the number of tourists are the optimal solutions sought.

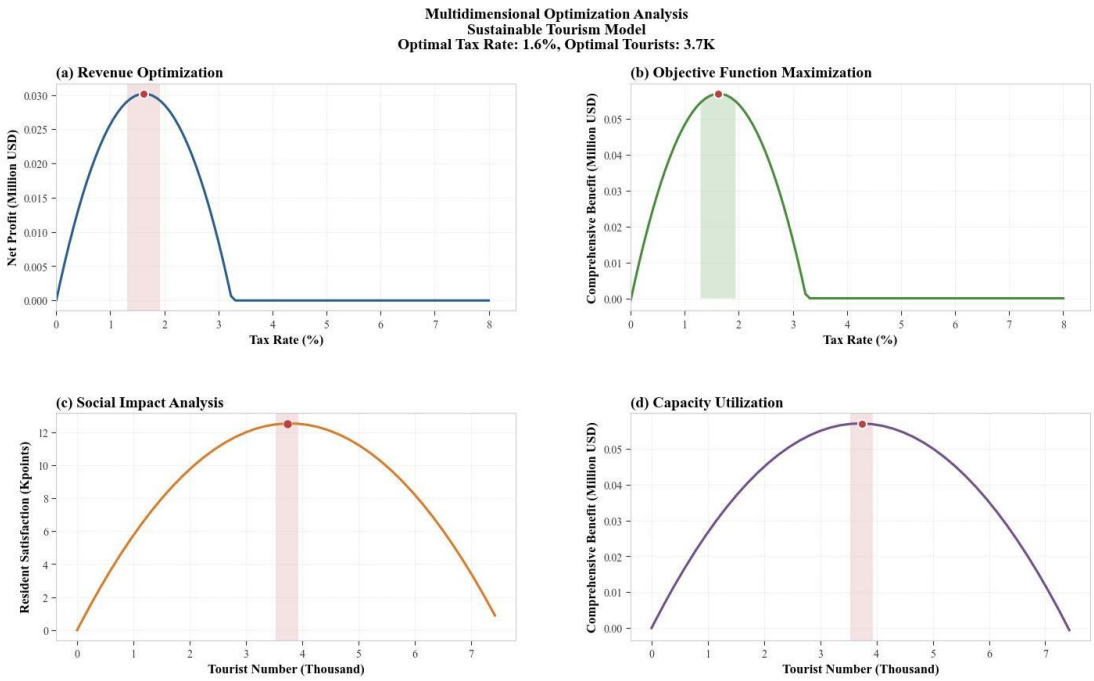


Fig13. Multidimensional optimization analysis

Fig13 reflects the two-dimensional relationship between tax rate-net income, tax rate-comprehensive income, number of visitors-residents' satisfaction, and number of visitors-comprehensive income.

4.1.9 Sensitivity analysis

In order to more clearly reflect which factors have a greater impact under the sustainable tourism model of the city of Juneau, we have conducted a sensitivity analysis.

Assume that the weights of the four key factors—the number of tourists$N(t)$, the rate of glacier

melt$G(t)$, environmental pressure$E(t)$and resident satisfaction$S(t)$—x_1、x_2、x_3、$x_4$are each equal to 1. The perturbation synthesis index I_{param} for each parameter can then be calculated by weighting the change in value of the four key factors with their respective weightsx_1、x_2、x_3、x_4. Suppose that the change in value of each factor is calculated by perturbing each parameter by 10%.

If the parameter p is a key parameter, then the change in value after perturbation can be expressed by the following equation (4-10):

$$\Delta p = p_{new} - p_{original} \tag{4-10}$$

Among them, the new value after applying a positive perturbation:

$$p_{new} = p_{original} \times (1 + 10\%) \tag{4-11}$$

New value after applying negative perturbation:

$$p_{new} = p_{original} \times (1 + 10\%) \tag{4-12}$$

Thus, the value of the perturbation change corresponding to each parameter is calculated.

Assuming that the weighted values of the four key x_1、x_2、x_3、$x_4$influencing factors satisfy the following relationship:

$$x_1 + x_2 + x_3 + x_4 = 1 \tag{4-13}$$

These four weights represent the contribution of each influencing factor to the perturbation synthesis index. The perturbation synthesis index of a parameter can be expressed by the following equation:

$$I = x_1 \times \Delta p_1 + x_2 \times \Delta p_2 + x_3 \times \Delta p_3 + x_4 \times \Delta p_4 \tag{4-14}$$

Where, respectively, Δp_1、Δp_2、Δp_3、Δp_4represent the change values of the number of tourists, the rate of glacial melt, environmental pressure and resident satisfaction. Then, based on the given weight values, the comprehensive index of disturbance can be calculated.

The final sensitivity analysis results are as follows:

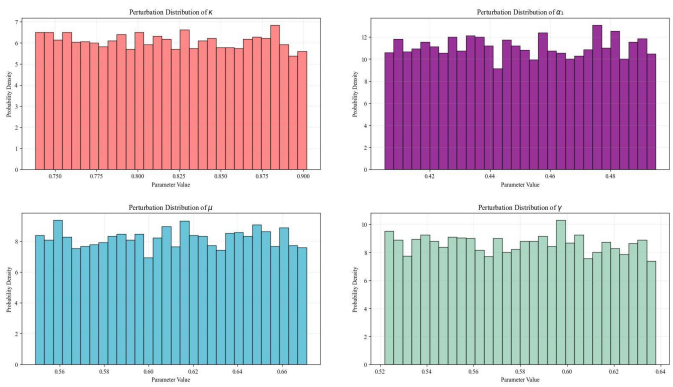


Fig14.Perturbation distribution of different parameters

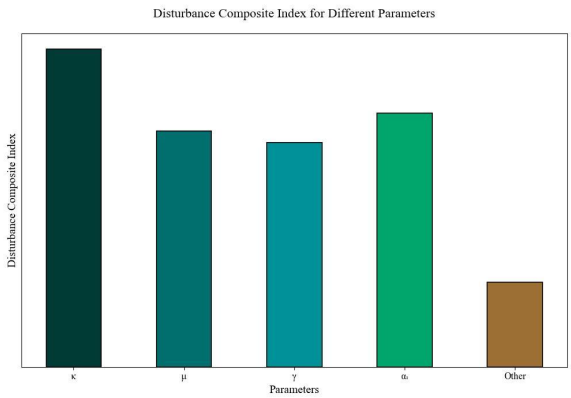


Fig15.Results of sensitivity analyses

From Fig15, the four parameters are ranked in order of importance as $\boldsymbol{\kappa > \alpha_1 > \mu > \gamma}$.

κ is expressed as constant factor for glacier melting, α_1 is expressed as the proportion of taxes and fees spent on the environment, μ is expressed as tourist growth rate and γ is expressed as tax impact on tourist number.

4.2 Application of sustainable tourism models

In order to apply the sustainable tourism model we have created to other tourist destinations affected by overtourism, we quantify each tourist destination using the three core dimensions of ecological sensitivity (E), cultural conservation (C) and infrastructure carrying capacity (I). The specific quantitative indicators are as follows:

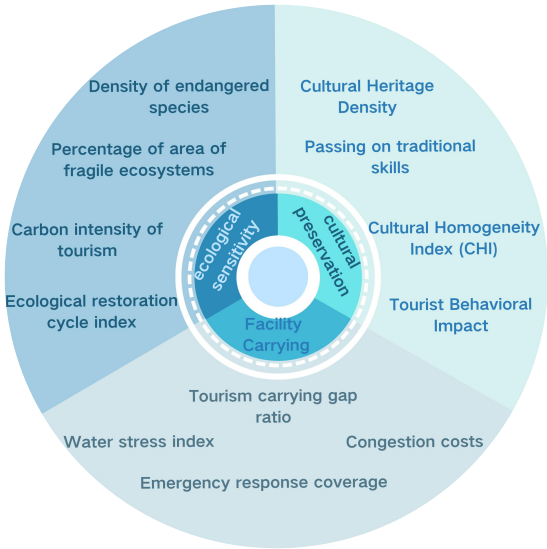


Fig16.Influencing factors at different dimensions（E、I、C）

4.2.1 Establishment of a core dimension calculation model

Raw indicator data is obtained for each dimension, but there is no guarantee that the collected data is all in the same dimension. To ensure the accuracy of the calculation results, the data is standardized to eliminate dimensional differences and process positive and negative indicators. The specific calculation is as follows:

Positive indicators (4-15):

$$x'_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \tag{4-15}$$

Negative indicators (4-16):

$$x'_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)} \tag{4-16}$$

Among them, the original data x_{ij}、$x_j$is the area of glacier degradation.

After obtaining the standardized data of each dimension, the weight of each indicator in each dimension is calculated using the entropy weight method, and the specific calculation is as follows Equation:

Calculate the entropy value $e_j$of the j indicator (4-17):

$$e_j = -\frac{1}{\ln n}\sum_{i=1}^{n} p_{ij}\ln p_{ij} \tag{4-17}$$

Among them， $p_{ij} = \frac{x'_{ij}}{\sum_{i=1}^{n} x'_{ij}}$

Calculation weight (4–18):

$$w_j = \frac{1-e_j}{\sum_{j=1}^{m}(1-e_j)} \tag{4-18}$$

Then, the positive and negative indicators and weights obtained above are weighted and summed to obtain a comprehensive score for each dimension (E, C, I). The specific calculation formula (4-19) is as follows:

$$E(C、I) = \sum_{j=1}^{m} w_j \times x'_{ij} \tag{4-19}$$

Finally, the calculation of the core dimensions is completed based on the score and weight combination.

4.2.2 Modeling of area type determination

In order to determine which type of region a region belongs to, it is necessary to define a construction type space *t* using mathematical statistical ideas in the type space for the determination of the type of region, set the three-dimensional eigenvectors (4-20):

$$x = [E, I, C]^T \in R^3 \tag{4-20}$$

Its basis vectors are normalised orthogonal dimensions.

When a region has three core dimensions, one of which satisfies the following equation (4-21), the region is judged to be of the type of region dominated by that core dimension:

For any *x*, there exists a unique principal type T_p ,such that:

$$T_p = arg\max_{k\in\{E,I,C\}}\left(\frac{z_k}{\sigma_{pooled}}\right) \tag{4-21}$$

where $\sigma_{pooled} = \sqrt{\frac{(n_E-1)\sigma_E^2+(n_I-1)\sigma_I^2+(n_C-1)\sigma_C^2}{n_E+n_I+n_C-3}}$

If the two remaining dimensions satisfy the following equation (4-22), the area is judged to be a composite of the two core dimensions, otherwise it is classified as a master type:

if and only if there exists $k1, k2 \in \{E, I, C\}$:

$$\frac{|z_{k_1} - z_{k_2}|}{\sqrt{\sigma_{k_1}^2 + \sigma_{k_2}^2}} < t_{\gamma_2}(n_{k_1} + n_{k_2} - 2) \tag{4-22}$$

Combining the data from the previous core dimension calculations, the following equation (4-23) is calculated:

$$CV = \frac{\sigma}{\mu} \times 100\% \tag{4-23}$$

where σ is the standard deviation of the four-dimensional scores and μ the mean of the four-dimensional scores.

If $CV \leq 20\%$, then the area is determined to be comprehensive; if $CV > 20\%$, then the area is determined to be non-comprehensive.

4.2.3 Classification of regional types

Based on the modelling of area type determination, we can classify area types into the following seven categories:

• Ecological sensitivity（E-dominant）

Regulated by the following parameters：

parametric	Rules	mathematical expression
$E_{\max}$	Decreases with increasing ecological score, nonlinearity suppressed	$E_{\max}^{new} = E_{\max}^{base} \times \left(1 - 0.3 \times \frac{E}{1.2}\right)$
β_1	Environmental investment efficiency accelerates as eco-scores improve	$\beta_1^{new} = \beta_1^{base} \times (1 + 0.5 \times \tanh(3E))$
κ	Glacier ablation rate positively correlates with ecological score	$\kappa^{new} = \kappa^{base} \times (1 + 0.2 \times E^2)$

• Cultural Preservation（C-dominant）

parametric	Rules	mathematical expression
$S_{\min}$	Satisfaction Lower Limit and Culture Scores are increasing in segments	$S_{\min}^{new}=\begin{cases} S_{\min}^{base}\times 1.2 & C\geqslant 0.8 \\ S_{\min}^{base}\times(1+0.25C) & C<0 \end{cases}$
η	Congestion loss coefficient decreases with increasing culture score	$\eta^{new}=\eta^{base}\times\left(1-0.15\times\sqrt{C}\right)$
λ	Employment Gains and Cultural Gains Exponential correlation between employment gains and culture scores	$\lambda^{new}=\lambda^{base}\times e^{0.3C}$

• Infrastructure Shortage (I-dominant)
Regulated by the following parameters：

parametric	Rules	mathematical expression
$I_{\max}$	Infrastructure stress ceilings dynamically adjusted with the carrying capacity score	$I_{\max}^{new}=I_{\max}^{base}\times(0.5+0.5\times I^{1.5})$
β_2	Infrastructure investment efficiency accelerates at low carrying capacity	$\beta_2^{new}=\beta_2^{base}\times(1+0.6\times(1-I))$
$N_{\max}$	Maximum visitor capacity has an S-shaped relationship with the carrying capacity score	$N_{\max}^{new}=N_{\max}^{base}\times\frac{1}{1+e^{-5(I-0.5)}}$

• Ecocultural (E-C type)
Regulated by the following parameters：

parametric	Rules	mathematical expression
$E_{\max}$	Ecological and cultural weighting mixed: adjustment $w_E=0.5, w_C=0.5$	$E_{\max}^{new}=0.5\times E_{\max}^{eco-rule}+0.5\times E_{\max}^{cultural\ rule}$
α_3	The proportion of social investment has increased, and the weight is related to *C* the score.	$\alpha_3^{new}=\alpha_3^{base}+0.1\times C$
κ	The rate of glacial melting is dominated by ecological scores, and cultural protection is indirect.	$\kappa^{new}=\kappa^{base}\times(1\times 0.2E-0.05C)$

• Infrastructure－Cultural (I-C model）

Regulated by the following parameters：

parametric	Rules	mathematical expression
β_2	The efficiency of investment in facilities increases exponentially with the score for infrastructure and culture.	$\beta_2^{new}=\beta_2^{base}*1.2^I*1.1^C$
α_3	The proportion of social project funds is linearly related to the culture score.	$\alpha_3=0.4+0.2C \quad (0.4\leqslant\alpha_3\leqslant 0.6)$
η	Congestion losses are reduced with comprehensive improvements in two dimensions.	$\eta^{new}=\eta^{base}\times 0.9^{(I+C)}$

• Ecological-Infrastructure (E-I)

Regulated by the following parameters：

parametric	Rules	mathematical expression
E	Upper pressure limit decreases with ecological score index.	$E_{\max}^{new}=E_{\max}^{base}\times 0.95^{3E}$
$N_{\max}$	The maximum visitor capacity is positively correlated with the facility score.	$N_{\max}=N_{\max}\times\frac{I}{0.6}$
η	Environmental efficiency is increasing rapidly with ecological pressure.	$\beta_1^{new}=\beta_1^{base}\times 1.3^E$

• Integrated

parametric	Rules	mathematical expression
$\alpha_i(t)$	Investment ratio matches current dimension weights in real time	$\alpha_i(t)=\frac{w_i(t)}{w_E(t)+w_I(t)+w_C(t)}$
$N_{\max}$	The upper limit for visitors is the lesser of 120 per cent of the reference value and the reciprocal of the pressure.	$N_{\max}=\min\left(1.2N_{base},\frac{1.5}{\max(E,I,C)}\right)$

Additionally, there also exist ecological-cultural(E-C), infrastructural-cultural (I-C), and ecological-infrastructural (E-I).

4.2.4 Modeling of attractiveness of tourist destinations

To better quantify tourism planning for a region, we need to model the impact of different policies (e.g. adjusting fares, limiting tourist arrivals, building transport facilities) on the attractiveness of attractions.From this, the attraction function (4-24) is introduced:.

$$A_j(t) = \frac{\beta_1 \alpha_1 R_j(t)}{1 + E_j(t)} \ln\left(1 + \frac{V_j}{V_{\max} - V_j}\right) e^{-0.1d} \tag{4-24}$$

where R_j is the fixed resource value of the attraction, V_j is the real-time number of tourists, E_j is an indicator of environmental stress, and d is the travelling time.

Based on the preliminary data research, it can be concluded that there are four main influences on the attractiveness of a place for tourism - resource value, number of tourists, environmental pressures, and travelling time.Then we analyse the interaction between different parameters separately:

It can be seen that attractiveness increases linearly with resource value R_j and inversely U-shaped with tourist arrival V_j ;environmental pressures $E_j$linearly reduce attractiveness; travelling time d non-linearly reduces attractiveness through the exponential decay term.

Increased tourist arrivals V_j increase attractiveness by $\ln\left(1 + \frac{V_j}{V_{\max} - V_j}\right)$,while environmental pressures inhibit attractiveness through $\frac{1}{1+E_j(t)}$;resource value R_j is directly proportional to attractiveness; and travelling time d non-linearly cancels out the advantage of resource value through the exponential decay term.

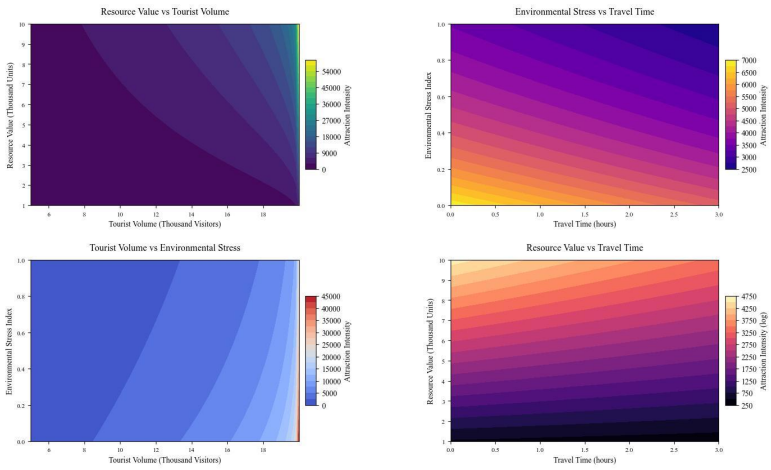


Fig17.The influence of the parameters of the attraction equation

4.2.5 Modeling of visitor flow control

In order to implement the core concepts of the sustainable tourism model at all times, it is crucial to balance the number of tourists with the pressure on the local environment.As a result, based on the establishment of our sustainable tourism model above, an exponentially weighted diversion algorithm based on capacity constraints is introduced to more easily determine whether the number of tourists is overloaded or not, and the specific solutions (4-25) are as follows:

Softmax function is applied to convert the attractiveness into a probability distribution:

$$Pi = \frac{e^{a_s}}{\sum_{j=1}^{n} e^{a_j}} \tag{4-25}$$

Where a_i is the attractiveness rating of attraction i.Plus the constraint treatment:

If $N_i > C_i$:

$$Overload: \Delta_i = N_i - C_i$$

$$\text{Remaining visitors } N_{remaining} = N_{total} - C_i$$

$$\text{reassign weight } \omega_j = \frac{P_j}{\sum_{k \neq i} P_k} (j \neq i)$$

Thus, we can determine whether the attraction is overloaded with tourists, and if it is overloaded, it can be deployed to another attraction in time to achieve dynamic balance, the following is an example:

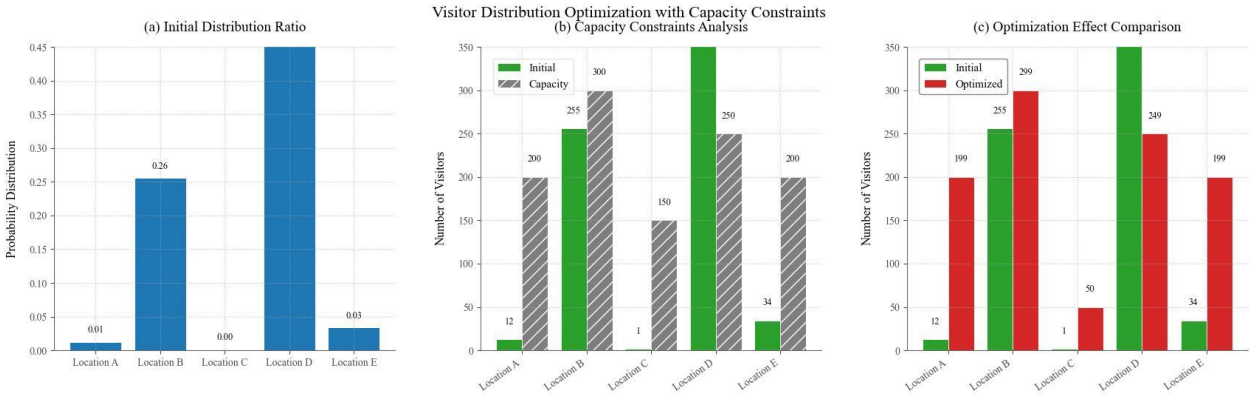


Fig18.Visitor distribution optimization with capacity constraints

4.2.6 Modeling of investment Feedback Mechanisms

Each attraction will have an original eco-compensation R_j that will be adjusted in proportion to the area of the attraction A_j and the total area of all attractions $\sum A_k$.This means that the amount of compensation transferred to each attraction $R_j^{transfer}$ will be adjusted according to the proportion of the area of the attraction to the total area of the attraction, so as to achieve a fair distribution of resources on the basis of the sustainable tourism model.The specific model formula (4-26) is as follows:

$$R_j^{transfer} = 0.2R_j \frac{A_j}{\sum A_k} \tag{4-26}$$

where $R_j^{transfer}$ is the amount of ecological compensation transferred to the site; R_j is the original ecological compensation of the site j; A_j is the area or resource volume of the site j; $\sum A_k$ is the sum of the area or resource volume of all sites; k represents all sites.

4.2.7 Results of the final model application

Based on the application and optimization of the sustainable tourism model, other cities were analyzed and the following results were obtained:

- **Ecological sensitivity (E dominated)**

Venice (Italy)

Environmental pressures: the lagoon ecosystem is fragile, and water quality has deteriorated due to pollution from cruise ships and tourist overload.

Other problems: aging infrastructure (e.g. bridges and roads), but residents are moving out due to high rents (only 10,000 local people remain), and cultural conflicts are relatively manageable.

Typical measures: large cruise ships are banned from entering the lagoon and a tourist entry fee is imposed.

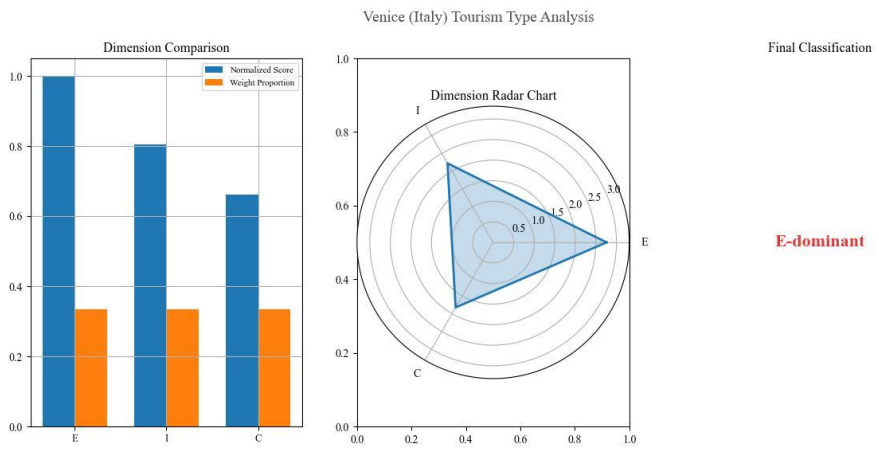


Fig19.Venice(Italy) tourism type analysis

Fig17 is a derivation of an example of a city classification model that analyzes Venice's metrics (environment, culture, infrastructure) to infer that Venice is an E-dominant city, i.e., that Venice needs to spend more time and effort on the environment.

- **Infrastructure shortages (I dominated)**

Mumbai (India)

Infrastructural pressures: classified as an “emerging member” city, traffic congestion, water scarcity, and overloaded tourist facilities.

Other issues: Residents' resistance to tourists is not yet a focal point, despite environmental hazards such as slums.

Context: Tourism is growing rapidly, but infrastructure renewal is lagging behind demand.

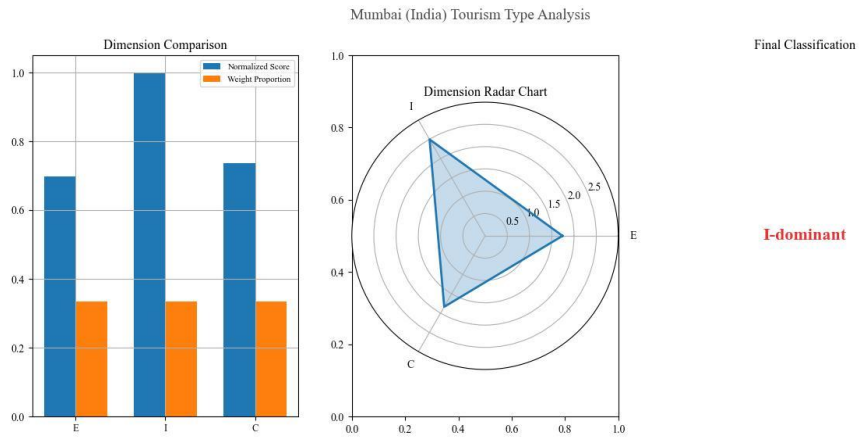


Fig20.Mumbai(India) tourism type analysis

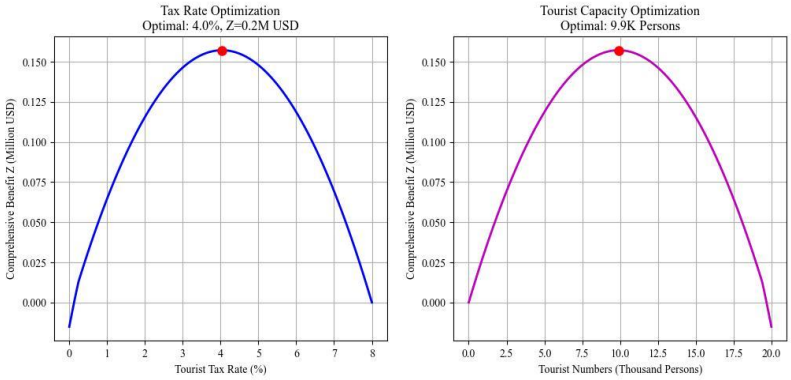


Fig21. Optimization programme in Mumbai

Fig18 is also a derivation of an example of a city classification model, which analyzes Mumbai's metrics to infer that Mumbai is an I-dominant city, i.e., that Mumbai's infrastructure overload is in dire need of being addressed.Fig19 is the optimal tax rate and optimal number of tourists for Mumbai obtained after analyzing Mumbai as an I-dominant city and after adapting the first-question constructed model to the city of Mumbai, which is 4.04% and 9,898 tourists per day, respectively.

- **Ecology-Infrastructure (E-I dominated)**

Reykjavik (Iceland)

Environmental issues: glaciers and geothermal landscapes are under ecological pressure due to the surge in tourists, and pollution issues are highlighted.

Infrastructural pressures: Airbnb has led to a housing crisis, with insufficient hotel capacity and a shortage of long-term rentals.

Residents' conflicts: although there has not been a large-scale outbreak, the rise in housing prices has triggered resentment among locals.

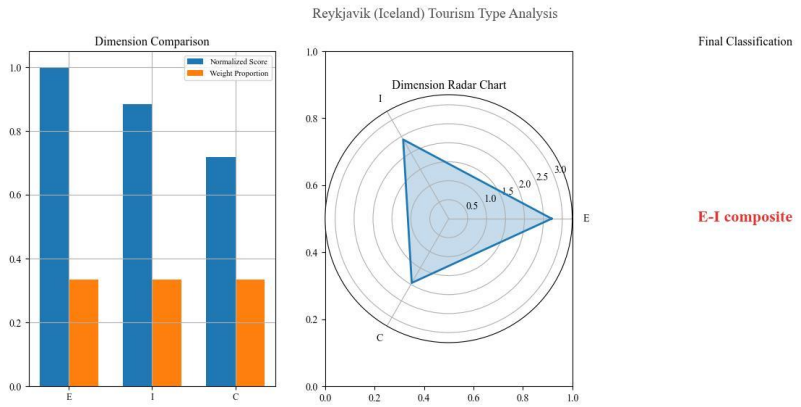


Fig22.Reykjavik(Iceland) tourism type analysis

V. Strengths and weaknesses

5.1 Strengths

• The sustainable tourism model can synthesize multifaceted factors, adapt to diversified objectives and constraints, and contribute to the development of sustainable tourism policies. The results of the model can be used in decision support systems.

• The regression model and discrete choice model used in the tourist attraction model can better reveal the laws of tourist behavior, which is a strong practical guidance for the evaluation of attractiveness and can provide strong support for marketing and product design.

• The analysis of regional types in this paper can help tourist destinations carry out market positioning, improve the efficiency of resource allocation and support targeted promotion. Through categorization, tourism operators can formulate more precise development strategies for different types of destinations.

5.2 Weaknesses

• Due to the multiple variables and objectives involved, the sustainable tourism models developed can be very complex and the solution process requires significant computational resources, especially in the optimization process. For uncertainties, model predictions may not be precise enough.

VI. Conclusions

• Sustainable Tourism Objectives: From the perspective of ecological, social and economic balance, it emphasizes the need for the tourism industry to take into account environmental protection and social and cultural heritage, and to promote its long-term development.

•Application of mathematical modeling: assessing the multiple impacts of policies and actions on the environment, resources and society with the help of methods such as multi-objective optimization and system dynamics to assist scientific decision-making.

• Environment and resource protection: simulate the ecological impact of tourism activities, formulate measures to reduce pollution and control consumption, and realize the coordinated

development of tourism growth and environmental protection.

• Social and cultural heritage: Assess the impact of tourism on community and cultural heritage to ensure that economic development does not destroy local culture and social structure.

• Future development prospects: With the advancement of data technology, mathematical modeling will help the tourism industry to move forward in the direction of green, smart and sustainable.

MEMORANDUM

KEY PREDICTIONS&EFFECTS&RECOMMENDATIONS

TO:
Juneau Tourism Council

FROM:
Sustainable Tourism Advisor

This memo outlines key predictions, effects of current measures, and recommendations to optimize Juneau' s tourism strategy for long-term sustainability and benefits.

OPTIMIZING JUNEAU'S TOURISM STRATEGY FOR A SUSTAINABLE FUTURE

STATUS QUO TRENDS POSE CHALLENGES

Environmental Pressures: Glacier melt projected to accelerate by 50% in 10 years,threatening tourism attractiveness due to overdependence on natural attractions and unmitigated emissions.
Community Burden: Short-term rentals projected to increase housing costs by 10-15%,straining infrastructure and potentially causing resident dissatisfaction.
Infrastructure Stress: Tourist-related maintenance costs may increase by 20-25% in 5 years. Municipal systems are strained: summer water use peaks at 3x residential demand, sewage overloaded by 40%, rubbish removal doubled in 5 years. Potential regional paralysis within 3 years without upgrades.

OPTIMIZATION RECOMMENDATIONS FOR ENHANCED OUTCOMES

TO MAXIMIZE BENEFITS AND MITIGATE NEGATIVE IMPACTS, I RECOMMEND

Dynamic Visitor Management: Use real-time data and reservations to distribute visitors to secondary/off-season attractions, optimizing resource allocation.
Transparent Revenue & Community Return: Publicly communicate revenue use and establish dedicated fund for conservation, infrastructure, resident benefits to build trust. Community
Community Participation & Governance: Create tourism-funded community development fund for local jobs. Involve residents in planning to foster positive perceptions.
Prioritize Environmental Investment: Expand investment in glacier/ecosystem restoration. Promote eco-friendly behaviors with public campaigns and green incentives.

POSITIVE MARKET TRENDS

Sustainable Tourism Demand: 60% of global travelers prefer sustainable tourism; eco-tourism market growing 8% annually.
High-Value Tourism: Sustainability attracts higher-spending tourists, enhancing competitiveness and revenue.

EFFECTS OF CURRENT MEASURES

Visitor Fees & Hotel Taxes: Projected $1.52M annual revenue for conservation and upgrades. Transparency crucial for public support.
Visitor Caps: 3762 daily cap at core attractions expected to reduce congestion by 21.3% and improve experience. Diversion strategies needed.
Green Transportation & Infrastructure: Investments projected to reduce peak traffic by 29.3% and lower carbon emissions.
Ecological Restoration Investment: about 40% of tax revenues invested in ecological restoration. Projected 15.7% tourism revenue increase and $2M annual economic benefits, reducing disaster risks and infrastructure costs.

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